Simulation Scope Module (SSCM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: SIMULOS® — Simulation Governance Architecture
Parent Standard: Simulation Purpose & Scope Standard (SPSS)
Operational Layer: Simulation Purpose & Scope Governance Layer
Category: Governance & Enforcement
Subcategory: Simulation Governance
Type: Parent Standard Module
Governed Space: Simulation Scope
Version: 1.0
Status: Canonical · Open Module
Effective Date: 16 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Simulation Scope

Canonical Definition

Simulation Scope Module (SSCM) defines the governance conditions under which the exact extent of a simulation inquiry is established, bounded, decomposed, maintained, changed, and documented relative to the governed Simulation Object and Simulation Purpose.

SSCM determines which object characteristics, functions, behaviors, interactions, operating conditions, temporal ranges, spatial ranges, system states, and inquiry dimensions are included within the simulation and prevents simulation outputs from being interpreted beyond the scope actually governed and executed.

Operational Role

SSCM serves as the second internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The preceding Simulation Purpose governance establishes:

Why is the simulation being performed?

SSCM asks:

What exactly must the simulation cover in order to address that purpose?

The governed progression is:

Governed Simulation Purpose → Scope Requirement → Scope Dimensions → Scope Inclusion → Scope Boundary → Scope Structure → Scope Status → Governed Simulation Scope

SSCM transforms simulation purpose into a concrete inquiry perimeter capable of guiding downstream Reality Representation, Assumptions, Model & Environment, Inputs & Data, Scenario Design, Execution, and Output Governance.

Module Operational Space

SSCM governs: Simulation Scope definition, scope dimensions, functional scope, behavioral scope, component scope, system scope, interaction scope, operational scope, environmental scope relevance, spatial scope, temporal scope, population scope, mission-phase scope, condition scope, operating-mode scope, failure-mode scope, capability scope, autonomy scope, scope hierarchy, partial scope, composite scope, scope decomposition, scope aggregation, scope dependencies, scope interfaces, scope expansion, scope contraction, scope recomposition, scope change, scope ambiguity, scope conflict, scope status, scope continuity, and Simulation Scope traceability.

Module Function

SSCM exists because knowing why a simulation is being performed does not establish what the simulation actually covers.

A simulation can have a legitimate purpose and still fail methodologically because its scope is: undefined, too narrow, unintentionally broad, fragmented, internally inconsistent, misaligned with purpose, silently expanded, silently contracted, or represented more broadly than what was actually simulated.

SSCM therefore creates the governed relationship:

Simulation Purpose → Required Simulation Coverage

Its function is to ensure that downstream simulation design addresses the intended inquiry without allowing omitted or unmodeled conditions to disappear from visibility.

Simulation Scope

Simulation Scope is the governed extent of the simulation inquiry.

It defines which portions of the Simulation Object and which simulation-relevant dimensions are intended to be represented and examined.

Simulation Scope may be defined across several dimensions simultaneously.

For example:

Object: Autonomous vehicle Functional Scope: Perception and braking Operational Scope: Urban low-speed driving Environmental Scope: Daylight and rain Temporal Scope: Single maneuver sequence Interaction Scope: Vehicle–pedestrian interaction

These dimensions together form the governed Simulation Scope.

Scope Must Derive From Purpose

SIMULOS® establishes:

Simulation Scope should be derived from the governed Simulation Purpose rather than from whatever the available model happens to support.

Existing simulation tools may influence implementation feasibility.

They must not silently redefine the inquiry. Purpose–Scope Alignment

A Simulation Scope should be sufficiently broad and sufficiently detailed to address the governed Simulation Purpose.

A Purpose–Scope Gap exists where the purpose requires investigation of conditions not covered by the Simulation Scope.

For example:

Purpose: Evaluate autonomous-system behavior across emergency conditions.

Scope: Normal operating conditions only.

The scope is structurally insufficient for the stated purpose.

Scope Dimensions

Simulation Scope may include several independent dimensions.

These may include:

Object Dimension — which object components or systems are covered.

Functional Dimension — which functions are examined.

Behavioral Dimension — which behaviors are represented.

Operational Dimension — which operating modes or conditions apply.

Environmental Dimension — which surrounding conditions matter to the inquiry.

Spatial Dimension — where the simulation applies.

Temporal Dimension — when or for how long the simulation applies.

Population Dimension — which populations, users, actors, or entities are represented.

Interaction Dimension — which interactions among systems or actors are included.

Capability Dimension — which capabilities are covered.

A governed Simulation Scope may require several of these dimensions to be combined.

Functional Scope

Functional Scope identifies which functions of the Simulation Object are included in the inquiry.

A system may contain many functions while only a subset is simulated.

For example:

Aircraft Simulation Object

may have scope limited to:

Landing Guidance + Braking

rather than:

Complete Aircraft Operation

SSCM requires this distinction to remain visible.

Behavioral Scope

Behavioral Scope identifies which object behaviors are included.

This may include: nominal behavior, degraded behavior, failure behavior, autonomous behavior, human interaction, recovery behavior, emergency behavior, or selected adaptive behavior.

A simulation of normal behavior does not automatically establish behavior under abnormal conditions.

Component Scope

Complex systems may require explicit component-level scope.

A simulation may include:

Subsystems A, B, and C

while excluding:

Subsystem D

Even where the complete system remains the Simulation Object, the inquiry itself may be partial.

Interaction Scope

Many simulation outcomes depend upon interaction among multiple components, actors, systems, or environments.

Interaction Scope establishes which relationships are included.

Examples may include: robot–human interaction, vehicle–vehicle interaction, spacecraft–ground interaction, AI-agent–tool interaction, system–infrastructure interaction, or subsystem–subsystem interaction.

Omitting a material interaction may create a scope limitation even where both participating objects are individually represented.

Operational Scope

Operational Scope identifies the operating conditions or modes within which the simulation applies.

Examples include: normal operation, startup, shutdown, emergency mode, autonomous mode, supervised mode, degraded mode, high-load operation, low-resource operation, maintenance mode, or mission-specific operation.

Operating Envelope

Where applicable, SSCM may establish a governed Simulation Operating Envelope.

This defines the range of operating conditions covered by the simulation.

A simulation result established within one operating envelope must not automatically be generalized beyond it.

Spatial Scope

Spatial Scope defines the geographic, physical, or logical area covered by the simulation.

Examples may include: one manufacturing cell, one warehouse, one city, one orbital region, one network segment, one infrastructure zone, or one defined physical volume.

Temporal Scope

Temporal Scope establishes the relevant time dimension.

A simulation may examine: milliseconds, one operational cycle, one mission phase, one day, one year, long-term system evolution, or another governed period.

A short-duration simulation must not automatically support conclusions about long-term behavior.

Population Scope

Where simulation concerns humans, users, agents, organizations, or populations, SSCM may define which population is included.

Population Scope may distinguish: age groups, user classes, geographic groups, operator classes, expertise levels, organizational roles, behavioral profiles, or other simulation-relevant populations.

Capability Scope

A complex AI or autonomous system may possess multiple capabilities.

Simulation Scope may include only defined capabilities.

For example:

Perception + Navigation

may be included while:

Autonomous Tool Execution

remains outside scope.

This distinction must remain explicit.

Autonomy Scope

Where autonomy is material, Simulation Scope should identify the autonomy level represented.

A simulation of:

Human-Supervised Operation

must not automatically be treated as simulation of:

Fully Autonomous Operation

where system behavior materially differs.

Failure and Abnormal Condition Scope

Failure conditions may be inside or outside Simulation Scope.

Where the Simulation Purpose includes safety, robustness, resilience, or failure analysis, applicable failure modes should be explicitly governed.

The absence of failure scenarios cannot later be interpreted as successful simulation of failure behavior.

Partial Simulation Scope

SIMULOS® permits partial simulation.

A simulation does not need to model the complete object or every possible condition.

However:

Partial Simulation Scope must remain explicitly partial.

A narrow but correctly governed simulation may be methodologically stronger than an apparently comprehensive simulation with undefined boundaries.

Scope Hierarchy

Complex simulation programs may require hierarchical scope:

Program Scope

↓

System Scope

↓

Subsystem Scope

↓

Scenario Scope

↓

Execution Scope

SSCM preserves the relationship among these levels.

Lower-level scope must remain traceable to the higher-level inquiry it supports.

Scope Decomposition

A broad Simulation Scope may be decomposed into manageable subscopes.

For example:

Complete Mission Scope

may be decomposed into:

Launch

Transit

Orbital Operation

Docking

Return

Decomposition is legitimate when the relationship among the subscopes remains governed.

Scope Aggregation

Multiple partial simulations may later contribute to a broader simulation inquiry.

However:

Aggregation of partial scopes does not automatically establish complete scope coverage.

The combined scope must be assessed for: missing areas, overlaps, incompatible assumptions, boundary gaps, interaction gaps, and uncovered transitions.

Scope Coverage

Scope Coverage describes the extent to which planned or completed simulation activity addresses the governed Simulation Scope.

Possible coverage states may include:

Complete

Substantially Complete

Partial

Fragmented

Insufficient

or:

Undetermined

Coverage does not determine simulation validity.

It determines whether the intended inquiry has actually been represented across its defined scope.

Scope Gap

A Simulation Scope Gap exists where a material part of the governed Simulation Scope is not covered by planned or completed simulation activity.

Examples may include: unrepresented operating mode, missing interaction, missing population, missing mission phase, missing failure condition, missing capability, or missing temporal range.

Scope gaps must remain visible.

Scope Overlap

Multiple simulation activities may cover the same portion of scope.

Overlap is not inherently problematic.

It may provide: redundancy, comparative analysis, sensitivity testing, or independent representation.

However, duplicated coverage should not be confused with broader coverage.

Scope Dependency

One portion of Simulation Scope may depend upon another.

For example:

Emergency Recovery Scope

may depend upon behavior established during:

Failure Initiation Scope

SSCM preserves material scope dependencies to prevent isolated simulation fragments from being interpreted without required context.

Scope Interface

A Scope Interface exists where two separately governed scope regions meet.

Interfaces may occur between: mission phases, subsystem simulations, model domains, populations, operating modes, or sequential simulation stages.

Where interface behavior is material, SSCM should prevent it from falling between separately governed scope segments.

No Scope Inflation Rule

SIMULOS® establishes:

A simulation claim must not extend beyond the Simulation Scope actually governed and covered.

Therefore:

One Function ≠ Complete System

One Scenario ≠ Complete Operational Envelope

One Environment ≠ All Environments

One Population ≠ Universal Population

One Mission Phase ≠ Complete Mission

unless a governed basis establishes broader applicability.

No Precision-for-Scope Substitution Rule

A highly detailed simulation of one narrow domain does not compensate for missing scope elsewhere.

SIMULOS® establishes:

Simulation Depth and Precision cannot substitute for missing Simulation Scope.

No Execution-for-Scope Substitution Rule

Repeating the same simulation many times does not expand its scope.

More Runs ≠ Broader Scope

unless new governed conditions are actually introduced.

No Model-Capability Scope Rule

The capabilities of an available simulation model must not silently determine the governed Simulation Scope.

If the purpose requires conditions the model cannot represent, that limitation must remain visible.

Scope Expansion

Scope Expansion occurs when new material functions, conditions, objects, populations, environments, capabilities, interactions, or periods are added to the Simulation Scope.

Expansion may require reassessment of: Reality Representation, Assumptions, Model & Environment, Inputs & Data, Scenario Design, Execution, Output Governance, and Reality Correlation.

Scope Contraction

Scope Contraction removes material portions of the previously governed Simulation Scope.

Contraction may be legitimate where: purpose changes, constraints emerge, simulation is decomposed, or a narrower inquiry is intentionally adopted.

It must remain explicit.

Scope Recomposition

Scope Recomposition changes the structure of scope without necessarily making it strictly broader or narrower.

For example:

several subsystem scopes may be reorganized into interaction-based scopes.

The relationship to the original Simulation Purpose must remain traceable.

Scope Change Materiality

A change is material where it could alter: simulation meaning, representation requirements, assumptions, scenario requirements, outputs, interpretation, or downstream applicability.

Material changes should trigger downstream reassessment.

No Silent Scope Change Rule

Material Simulation Scope must not change without explicit governance.

This applies both before and after execution.

No Retrospective Scope Adjustment Rule

Simulation Scope must not be narrowed after results are known merely to remove failed or inconvenient conditions from the represented inquiry.

Likewise, it must not be expanded retrospectively to make favorable results appear more widely applicable.

Scope Ambiguity

Scope Ambiguity exists where it is unclear whether a material dimension belongs inside or outside the simulation inquiry.

Ambiguity may concern: functions, conditions, populations, interactions, operating modes, environments, or time periods.

Material ambiguity must remain explicit until resolved.

Scope Conflict

A Scope Conflict exists where credible simulation records or stakeholders define materially different intended scopes for the same simulation activity.

For example:

engineering may treat a simulation as:

Component Evaluation

while management presents it as:

System Safety Simulation

Such conflict must be resolved or explicitly preserved.

Scope Status

Possible Simulation Scope statuses may include:

Established

Conditionally Established

Partially Established

Scope Ambiguous

Scope Conflicted

or:

Scope Undetermined

Status describes the quality of scope definition, not the quality of simulation execution.

Scope Continuity

Simulation Scope should remain traceable through the full simulation lifecycle.

Conceptually:

Simulation Purpose → Governed Scope → Representation → Scenarios → Execution → Outputs

This allows later determination of whether a simulation output actually corresponds to the scope claimed for it.

Simulation Scope Record

SSCM should establish a persistent Simulation Scope Record capable of reconstructing: scope basis, scope dimensions, inclusions, functional coverage, behavioral coverage, operational conditions, environmental relevance, spatial range, temporal range, population, interactions, capabilities, partial scopes, dependencies, gaps, changes, and current scope status.

Failure Conditions

SSCM governance failure may exist where: Simulation Scope is not explicitly established, scope cannot be connected to the Simulation Purpose, material scope dimensions remain undefined, the scope is insufficient for the stated purpose, partial simulation is represented as complete coverage, material scope gaps are hidden, different scopes are aggregated without coverage analysis, scope interfaces remain ungoverned, scope is determined solely by available model capability, scope expands or contracts silently, retrospective scope adjustment occurs, materially different scopes are treated as equivalent, or simulation outputs are represented beyond the scope under which they were produced.

Minimum Implementation Framework

1. Derive Scope from the Simulation Purpose

Translate the governed Simulation Purpose and Simulation Questions into the required simulation coverage. 2. Define Material Scope Dimensions

Establish the relevant functional, behavioral, component, operational, environmental, spatial, temporal, population, interaction, capability, and other applicable scope dimensions. 3. Structure and Bound the Scope

Define complete, partial, composite, or decomposed scope and identify material dependencies, interfaces, and known gaps. 4. Govern Scope Change and Status

Record scope expansion, contraction, recomposition, ambiguity, conflict, and current Simulation Scope Status. 5. Preserve Scope Traceability

Create the Simulation Scope Record and maintain the relationship between purpose, scope, downstream simulation design, execution, and resulting outputs.

Governance Outputs

SSCM may produce: Simulation Scope Definition, Simulation Scope Record, Purpose–Scope Alignment Record, Simulation Scope Map, Scope Dimension Record, Functional Scope Record, Behavioral Scope Record, Component Scope Record, Interaction Scope Record, Operational Scope Record, Simulation Operating Envelope, Spatial Scope Record, Temporal Scope Record, Population Scope Record, Capability Scope Record, Autonomy Scope Record, Failure Condition Scope Record, Partial Scope Record, Composite Scope Record, Scope Hierarchy, Scope Decomposition Map, Scope Aggregation Assessment, Scope Coverage Assessment, Simulation Scope Gap Record, Scope Overlap Record, Scope Dependency Map, Scope Interface Record, Scope Expansion Record, Scope Contraction Record, Scope Recomposition Record, Scope Change Assessment, Scope Ambiguity Record, Scope Conflict Record, Simulation Scope Status, and Simulation Scope Traceability Map.

Use Case 1 — Autonomous Vehicle Operational Scope

Scenario

An autonomous vehicle developer intends to simulate pedestrian avoidance before urban deployment.

The Simulation Object is already identified and the purpose is:

Evaluate pedestrian-avoidance behavior under selected urban operating conditions. Application

SSCM establishes scope across:

Functional Scope: Perception, prediction, path planning, and braking.

Operational Scope: Urban driving below the defined speed threshold.

Environmental Scope: Daylight, dry conditions, and selected rain conditions.

Interaction Scope: Vehicle–pedestrian interaction.

Population Scope: Defined pedestrian behavior profiles.

Night-time operation, snow, highway driving, and vehicle-to-vehicle collision scenarios remain outside the governed Simulation Scope. Result

Successful simulation results cannot later be represented as evidence that the vehicle has been simulated across its complete autonomous-driving operational envelope. Use Case 2 — Integrated Space Mission Simulation

Scenario

A space organization intends to simulate an integrated mission from launch through orbital operation.

The initial mission program contains separate simulation environments for:

Launch

Orbital Insertion

Navigation

and:

Docking Application

SSCM decomposes the broader mission scope into controlled subscopes while identifying the interfaces between mission phases.

The architecture detects that the transition between:

Orbital Insertion

and:

Navigation

is not covered by either simulation environment.

This becomes a documented Simulation Scope Gap. Result

The organization cannot claim complete mission simulation merely because each major mission phase has an individual simulation.

The uncovered transition remains visible until governed simulation coverage is established.

Architectural Position

Simulation Scope Module (SSCM) is the second internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The internal progression begins:

Simulation Purpose → Simulation Scope

The preceding purpose module establishes:

Why is the simulation being performed?

SSCM establishes:

What exactly must that simulation cover?

The next SPSS governance spaces separately govern:

what must be explicitly excluded,

where the simulation is contextually applicable,

and:

how much simulation depth and precision the intended inquiry requires.

SSCM does not replace Validation Scope under VALIDOS®, Operational Reality governance under ORA™, Simulation Object Boundary governance under SOS, or Reality Representation governance under RRS.

It governs specifically:

the extent of the simulation inquiry itself.

Foundational Principle

A simulation can legitimately support only the portion of the intended inquiry that falls within the scope actually governed and covered by simulation.

Canonical Closing Statement

Simulation Scope Module (SSCM) establishes the governed extent of the simulation inquiry by translating Simulation Purpose into explicit functional, behavioral, component, interaction, operational, spatial, temporal, population, capability, and other materially relevant scope dimensions. Through purpose–scope alignment, partial-scope governance, decomposition, aggregation control, coverage assessment, gap detection, interface governance, anti-inflation rules, change control, scope status, and complete traceability, SSCM prevents narrow or fragmented simulation activity from being represented as broader coverage than was actually established. By defining exactly what the simulation must cover before downstream representation, assumptions, modeling, scenario design, execution, and output interpretation proceed, SSCM provides the second internal governance mechanism of the Simulation Purpose & Scope Standard.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

https://originopenfoundation.org/